

Computer Organization and Architecture		
Lab #:	09	
Lab Title:	Palindrome Number	
Submitted by:		
Names		Registration #
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Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Palindrome Number

Task: use (Use Xilinx):

(Define input and output as vectors ; and not multiple variables)

- Write the Verilog gate-level description and testbench for a circuit with 3-bit input(X) and 1 bit output (Y). The output is 1 if the input number is a palindrome (reads the same backward and forward eg 101 or 010). The minterms of the output will be.. (hint: by gate level not using assignment operator).

```
module palindrome(input [2:0]x,output y);
```

Palindrome Verilog:

```
module palindrome (input [2:0] X,      // Inputs of UUT decided as ?reg?
output Y);      //Outputs of UUT (is this case module verilog?) decided as // wire

wire X0_NOT, X1_NOT, X2_NOT, AND1, AND2, AND3, AND4, OR1;
not(X0_NOT, X0);
not(X1_NOT, X1);
not(X2_NOT, X2);

and(AND1, X2, X0);      // Implement the logic using AND and OR gates
and(AND2, X1_NOT, X0_NOT);
and(AND3, X2_NOT, X1);
and(AND4, X2, X1, X0);
or(Y, AND1, AND2, AND3, AND4);
endmodule
```

TestBench:

```
module palindrome_tb;
reg [2:0] X;          // Inputs of UUT decided as ?reg?
wire Y;              //Outputs of UUT (is this case module verilog?) decided as // wire

palindrome UUT(X, Y);
initial begin
$monitor("X=%b, Y=%b", X, Y);
    // Test cases
    X = 3'b000; #10;
    X = 3'b001; #10;
    X = 3'b010; #10;
    X = 3'b011; #10;
    X = 3'b100; #10;
    X = 3'b101; #10;
    X = 3'b110; #10;
    X = 3'b111; #10;
    $finish;
end
endmodule
```

Result:

